

MATERIALS SCIENCE

Influence of voids on the thermal and light stability of perovskite solar cells

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The formation of voids in perovskite films close to the buried interface has been reported during film deposition. These voids are thought to limit the efficiency and stability of perovskite solar cells (PSCs). Here, we studied the voids formed during operation in perovskite films that were optimized during the solution deposition process to avoid voids. New voids formed during operation are found to assemble along grain boundaries at the bottom interface, caused by the loss of residual solvent and conversion of amorphous phase to crystalline phase. Unexpectedly, the formation of these voids did not negatively affect the stability of PSCs. Decreasing the amorphous region in perovskites by thermal annealing decreased the positive iodide interstitial density, and improved the light stability of PSCs. The annealed devices maintained 90% of their initial efficiency and light soaking for 1900 hours at open circuit condition under 1-sun illumination at 50°C.

INTRODUCTION

Perovskite solar cells (PSCs) have witnessed a tremendously fast development in recent years, and the certificated power conversion efficiencies (PCEs) have exceeded 25% for small-area single-junction cells and nearly 30% for tandem cells (1, 2). Scalable solution-based deposition methods have been developed to fabricate large-area perovskite minimodules in research laboratories and modules in industry (3, 4). The efficiencies of perovskite modules are also quickly rising with improved controlling of material uniformity of perovskite and other active layers (5, 6). However, the instability of PSCs under operation conditions is still a major challenge to be addressed before the commercialization of PSCs. The best-performing PSCs contain hole transport layer, perovskites, electron transport layer, and conductive electrodes. Almost every active layer and their interfaces cause instability issues, while the perovskite layer itself still appear to be the weakest one (7, 8).

Tremendous efforts have been devoted to enhancing the intrinsic property of perovskites. Methylammonium ions (MA^+) are frequently reported to induce quick degradation of perovskites, particularly those tin-based perovskites (9, 10). Reducing its content to less than 5% or eliminating it from perovskite composition, i.e., formamidinium-cesium mixed-cation lead triiodide ($\text{FA}_x\text{Cs}_{1-x}\text{PbI}_3$), was shown to markedly improve the perovskite stability (3). Nevertheless, other element-related defects, such as iodide defects, are still present in MA^+ -free perovskites and can cause fast degradation of perovskites (11, 12). We showed that a defect compensation using slightly excess amount of formamidinium iodide (FAI) or cesium iodide (CsI) in $\text{FA}_x\text{Cs}_{1-x}\text{PbI}_3$ can reduce the initial iodide interstitial concentration, which delays the degradation of the perovskite under operation conditions (3). Because the degradation of perovskites generally starts from defective sites such as surfaces and interfaces, even more studies have been done to protect the top surface of perovskite films, either passivating the surface defects and/or convert the surface defective layer into robust inorganic materials or layered perovskites (13, 14). Defects at the imbedded bottom interfaces have received much less

attention in the past mainly because it is not as straightforward to monitor or modify this interface, unlike surface treatment. We recently observed the formation of large number of three-dimensional voids at the imbedded interface of perovskites and hole transport layer in p-i-n-structured solar cells during the film formation stage (15). These voids randomly distribute at the bottom interface with size ranging from 50 to 300 nm. They are formed because of the evaporation of the entrapped nonvolatile solvent additives such as dimethyl sulfoxide (DMSO) during the film annealing stage. Perovskite devices with more such voids are generally less efficient and stable under illumination.

Here, we studied the void formation and its impact on device stability during operation even if the as-fabricated perovskite films do not have voids. The voids showed different distribution from those formed during film fabrication. We also studied influence of these voids on thermal and operation stability of PSCs by manipulating the void formation, which also showed different but unexpected promising results of stabilizing perovskites.

RESULTS

Instead of using $\text{MA}_{0.3}\text{FA}_{0.7}\text{PbI}_3$, we chose $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ for this study because of its much better intrinsic stability, as reported previously (3, 9). We observed similar void formation during perovskite film fabrication for this composition as long as the solvent contained DMSO. Using the previously demonstrated additives, we controlled the solvent-drying process using the appropriate amount of DMSO and carbonyldiimidazole (CDI) so that the bottom interfaces do not have notable voids right after fabrication (15). We then studied the morphology changes of perovskite film bottom interfaces near tin-doped indium oxide (ITO) substrates using scanning electron microscope (SEM) by peeling off the blade-coated $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ films from the ITO glass substrates. To simulate the perovskite degradation in real devices, we fabricated perovskite devices for this study, which had a p-i-n structure of ITO/poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine (PTAA)]/ $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ /C₆₀/bathocuproine (BCP)/copper (Cu). PTAA and perovskite layers were blade-coated at room temperature in air. C₆₀ and Cu were deposited by the evaporation method. BCP was spin-coated onto the C₆₀ layer. Epoxy were used as encapsulant to cover the perovskite devices for stability test in air.

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The peeling off process is illustrated in Fig. 1A, where the epoxy strongly bonded to perovskite top surface so that perovskite films can be peeled off from the bottom perovskite/PTAA interface. The SEM images in Fig. 1 (C and D) show that the fresh perovskite-substrate interfaces are compact, and there was almost no void along the grain boundaries. However, the voids could still form and grow at the bottom interface for the perovskite films either under heat treatment at 75°C or under illumination with light intensity of 1 sun, which also heated the devices to 50°C. These two stability testing conditions were chosen to distinguish the influence of heat and light, which cause different degradation processes of PSCs. Light soaking can induce additional degradation paths, such as iodine formation, which was

reported to accelerate the degradation of perovskites (11, 16, 17). Figure S1 shows more top-view SEM images of perovskite-substrate interfaces in devices annealed or light-soaked for different durations. It can be seen from the SEM images in Fig. 1 (C and D) and fig. S1 that voids grew from several nanometers up to ~300 nm after thermal annealing or light soaking for 1000 hours. The voids at the interface were initially too small to be observed by cross-sectional SEM (Fig. 1E). After annealing for 520 hours, the voids were large and dense enough so that they can be observed in almost every cross-sectional SEM image. We did statistics on the void change, which is characterized by their diameters and the percentage of void volume compared to the total film area. As shown by Fig. 1F, voids

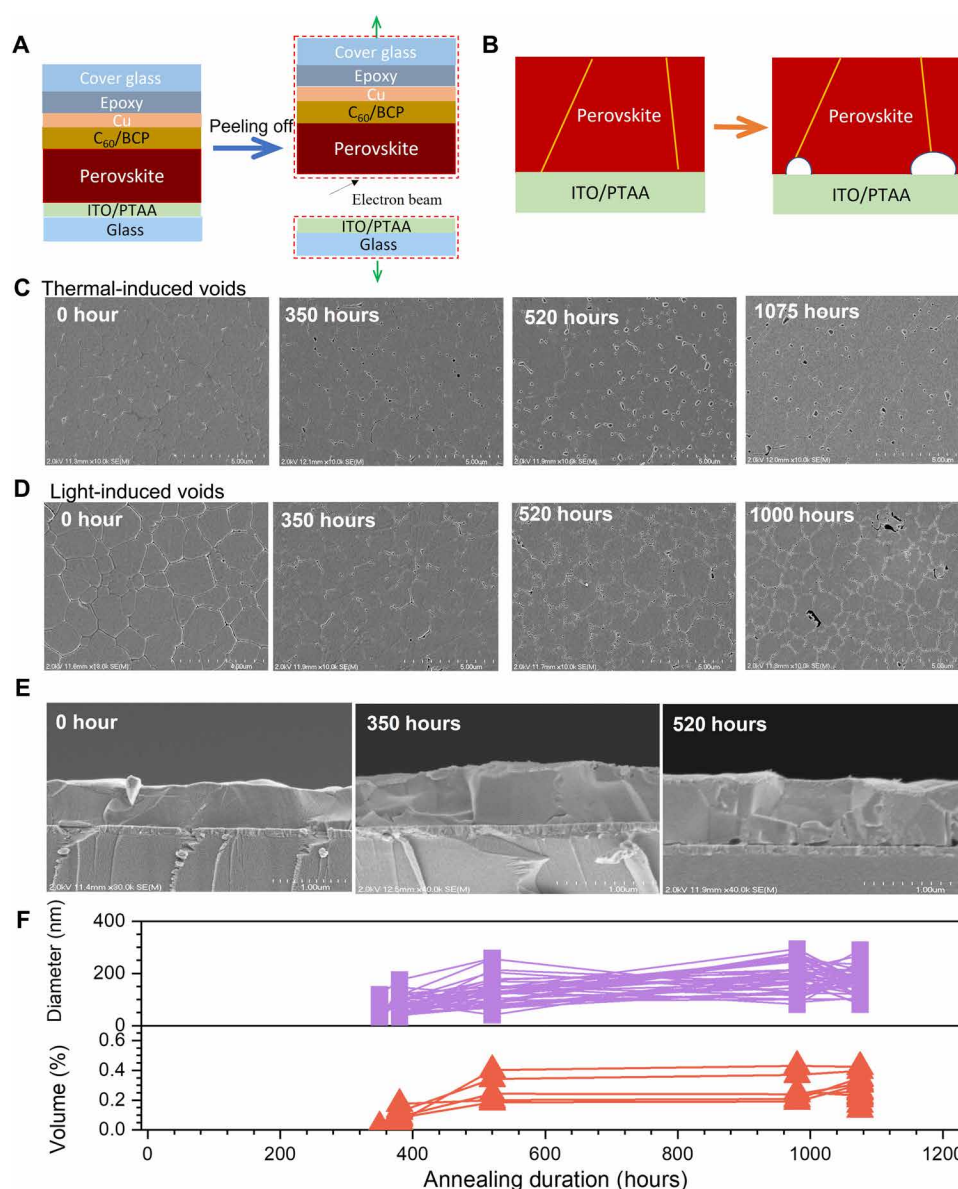


Fig. 1. Voids generation and volume statistics. (A) Schematic illustration of the peeling off perovskite films from ITO glass substrates with an epoxy encapsulant for SEM characterization. (B) Schematic illustration of voids location at the perovskite/substrate interface. (C) Top-view SEM images of perovskite-substrate interfaces of PSCs annealed at 75°C for different durations in dark. (D) Top-view SEM images of perovskite-substrate interfaces of PSCs light-soaked under 1 sun for different durations at the temperature of ~50°C. (E) Cross-sectional SEM images of PSCs after annealing at 75°C for different durations. (F) Voids size and percentage of void volume to perovskite layer after the devices were thermal annealed at 75°C for different durations.

started to appear after annealing for 320 hours, with average size increased from almost 0 to an average of ~ 150 nm after annealing for 520 hours. After that, the void size or volume percentage kept almost unchanged with further extended annealing. We verified the void formation using cross-sectional SEM images of devices. The average void volume is about 0.25% to perovskite layers after 520 hours. The void volume ratio was estimated by multiplying the average area ratio of voids by the void average depth ratio to film thickness ($\sim 100/800$ nm). The volume percentage of voids was calculated on the basis of voids area shown in SEM images with details in the Supplementary Materials (fig. S2). The voids formed during operation stage have very different morphology with those formed during film drying process (15). Almost all these voids were located at the grain boundaries close to perovskite-substrate interface, as illustrated in Fig. 1B. In addition, the size of voids even after annealing or light soaking for 1000 hours was still much smaller than grain size.

As shown in fig. S3, void formation during the aging process was also observed in $\text{MA}_{0.6}\text{FA}_{0.4}\text{PbI}_3$ solar cells after the devices were annealed at 75°C in dark for about 500 hours. We examined several possible mechanisms for the void formation based on the location of the voids. Like the void formation during film fabrication, it is possible that the residual DMSO in films along grain boundaries evaporates during annealing or long-term operation and thus leaves the voids behind. DMSO is still needed in our perovskite deposition to enhance the crystallinity of perovskites. Because of the high boiling point (189°C) of DMSO, we speculated that a small amount of residual DMSO may be still left in the perovskite films even after annealing at 150°C for 3 min, and the removal of residual DMSO during the long-term stability test would leave voids at the perovskite-substrate interfaces, as illustrated in Fig. 2A. This is confirmed by a control study that the perovskite films fabricated by the two-step process without DMSO did not show voids at perovskite/substrate interface after similar annealing conditions (fig. S4A), while voids formed for the perovskite films fabricated by the two-step process with DMSO (fig. S4B). Proton nuclear magnetic resonance (^1H NMR) characterization was conducted to quantify residual DMSO amount in $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ by comparing the integrated area of DMSO ^1H NMR peak and those of FA. As shown in fig. S5, the annealing process during the fabrication of perovskite films could not completely remove the DMSO solvent. There was still about 0.11 mole percent (mol %)

DMSO (to Pb) trapped in $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ films even after we elongated the annealing duration to 10 min. After the perovskite films were post-annealed at 75°C for 500 hours in the dark, the residual DMSO was reduced to 0.04 mol %. The evaporation of residual DMSO in perovskite layers could induce the volume shrinkage of perovskite layers. We then calculated the volume shrinkage using the lattice volume of intermediate phase $\text{DMSO}\cdot\text{PbI}_2\cdot\text{FAI}$ (CsI) and black phase $\text{FA}_x\text{Cs}_{1-x}\text{PbI}_3$. The lattice volume of intermediate phase is about $931 \text{ \AA}^3$ and that of $\text{FA}_x\text{Cs}_{1-x}\text{PbI}_3$ is about $256 \text{ \AA}^3$ (18–20). The volume shrinkage is $\sim 73\%$ when the intermediate phase is converted to $\text{FA}_x\text{Cs}_{1-x}\text{PbI}_3$. Consequently, the 0.07 mol % DMSO removal could only cause 0.05% volume shrinkage of perovskite films, which concludes that the escape of DMSO only contributed about 20% of formed voids during thermal annealing or device operation.

The perovskite decomposition from $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ to lead iodide (PbI_2) can also potentially reduce the film volume, but the phase transition from the black phase to the yellow phase increases the volume of films, as illustrated in Fig. 2 (B and C). However, x-ray diffraction (XRD) results in fig. S6A show that XRD peak intensity barely changed after the films were annealed at 75°C for 500 hours, and no additional peaks related to PbI_2 or yellow phase appeared. The XRD results indicate that perovskite decomposition and phase transition from the black to the yellow phase are not the reasons for the formation of the voids. In addition, the post-annealing process or light soaking process could induce grain growth and the conversion of amorphous phase material into crystalline phase due to solid-state diffusion of ions under thermal or photothermal conditions (21), which would also reduce perovskite volume, as illustrated in Fig. 2D. Previous studies using high-resolution tunneling electron microscope and other methods revealed the presence of nonnegligible amount of amorphous phase perovskites in films that yield very high efficiency (22–24). The solid-state ion diffusion under long-term operation under light or heat may cause the grain growth and conversion of these amorphous phases to the crystalline phases. We did observe a slight increase of average grain size from 1.14 to $1.37 \mu\text{m}$ after the films were annealed for 1000 hours, as shown in Fig. 3. Last, considering that strain relaxation may also cause material volume change, we examined the strain in out-of-plane direction before and after the film was thermally annealed for 500 hours. As shown in fig. S6B, the (001) and (002) diffraction peaks of the perovskite films did

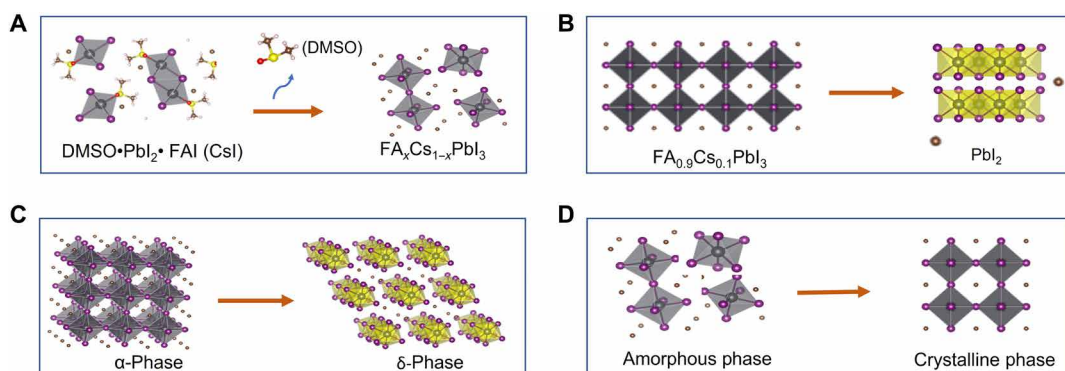


Fig. 2. Schematic illustration of perovskite volume variation by different processes. (A) Voids form because of the releasing of pretrapped DMSO during device operation. (B) Perovskite decomposition to PbI_2 decreases the volume of perovskite layers. (C) Phase transition from the black perovskite phase (α -phase) to the hexagonal nonperovskite phase (δ -phase) increases the volume of perovskite layers. (D) The conversion of the amorphous phase perovskite to the crystalline phase during recrystallization at thermal annealing causes the volume shrinkage of perovskite layers to form voids in perovskite layers.

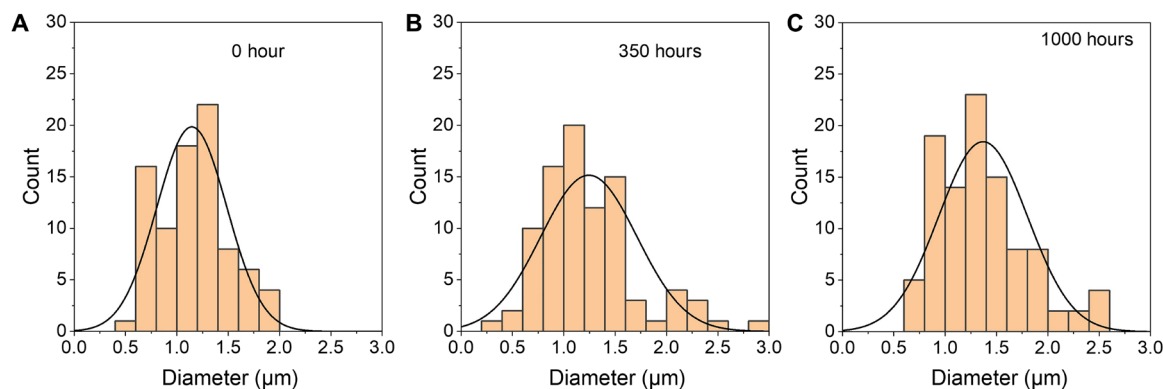


Fig. 3. Grain growth during stability testing. The distribution of perovskite grains in fresh perovskite films (A), and the perovskites films annealed at 75°C in dark for 350 hours (B) and annealed at 75°C in dark for 1000 hours (C). The average grain size increased from 1.14 to 1.24 μm and 1.37 μm after annealing for 350 and 1000 hours, respectively.

not show obvious change during voids generation, indicating that strain is not related to the volume shrinkage in $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$. This analysis concludes that the phase transition from amorphous to crystalline phase is likely the main reason for void formation.

We first studied the impact of voids on the thermal stability of PSCs by annealing the encapsulated $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ devices at 75°C in the dark and measured the PCE evolution at different post-annealing durations. For comparison, we plotted the evolution of void volume percentage in the same figure. As shown in Fig. 4A, the average PCE of PSCs decreased by 10% of the initial PCE after annealing for about 1100 hours at 75°C tested from seven devices fabricated from different batches. However, there was barely any step-like efficiency loss during the void formation period (annealing for 350 to 520 hours). The evolution of current density–voltage (J - V) curves and photovoltaic parameters of one solar cell that annealed at 75°C in the dark for different durations were shown in fig. S7 and table S1, respectively. In addition, the device efficiency after annealing in the dark could recover and even increase by 10% when the device was exposed to light (fig. S7), indicating that the post-annealing process did not actually damage the $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ perovskite. Therefore, we conclude that the voids formed during long-term annealing do not really cause the degradation of perovskites.

We then investigated the influence of voids on the stability of devices under illumination at open circuit (V_{OC}) condition, which is a harsher condition than maximum power point tracking (25, 26). This stability test is different from thermal stability because the generation of I_2 mainly occurs when the perovskites are exposed to light (15, 27). We have a concern that the formed voids could be a reservoir of formed I_2 , which may further accelerate the degradation of bottom perovskites. The perovskite devices were first post-annealed at 75°C in the dark for about 500 hours to generate enough voids in the perovskite layer, and then, they were soaked under white light with 1-sun intensity in air, which also heat up the devices to 50°C. The efficiencies of the devices over different light soaking durations were shown in Fig. 4B. As a control, the light stability of devices without post-annealing process was also shown. It can be seen that the devices with preformed voids had an even slower efficiency degradation rate, indicating that the post-annealing process actually enhanced the light stability of the perovskite devices despite of the formation of voids. The best T_{90} (T_{90} is the time over which the device efficiency reduces to 90% of its initial efficiency) of the devices

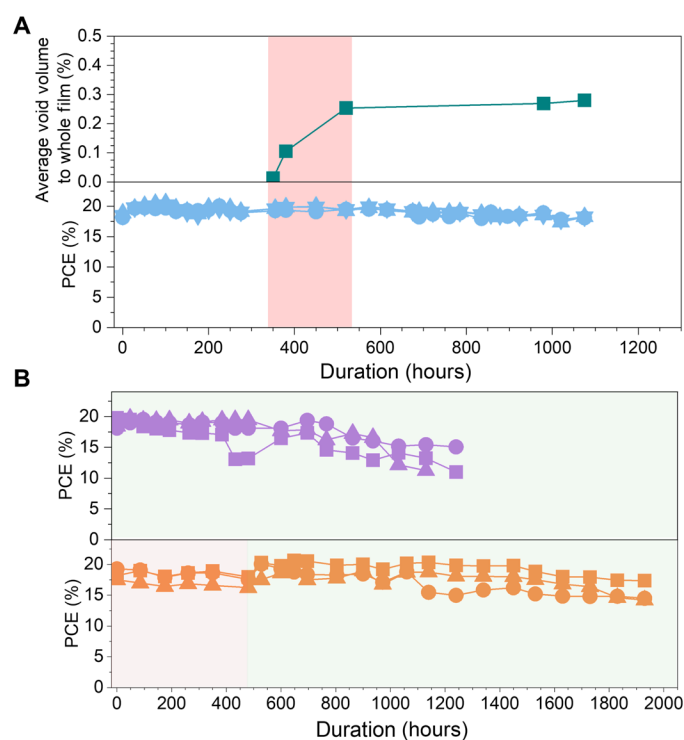


Fig. 4. Stability of PSCs. (A) The average percentage of void volume to perovskite layer (top) and efficiency evolution (bottom) of encapsulated PSCs after thermal annealing at 75°C for different duration. (B) Efficiency evolution of encapsulated PSCs under white light with an intensity of 1-sun illumination at V_{OC} condition (top); efficiency evolution of encapsulated PSCs that post-annealed at 75°C in dark for 480 hours and then under 1-sun illumination at V_{OC} condition (bottom).

was ~ 1900 hours, which is much better than devices that were light-soaked directly after fabrication ($T_{90} \sim 800$ hours). This is also among the best photostability for PSCs measured under operation conditions. The J - V curves of solar cells with and without preformed voids under light soaking for different durations and their photovoltaic parameters are shown in fig. S8 and table S2, respectively.

Because the formation of voids is not responsible for the degradation of the solar cells, we conducted defect density measurement to understand what caused the degradation of devices under

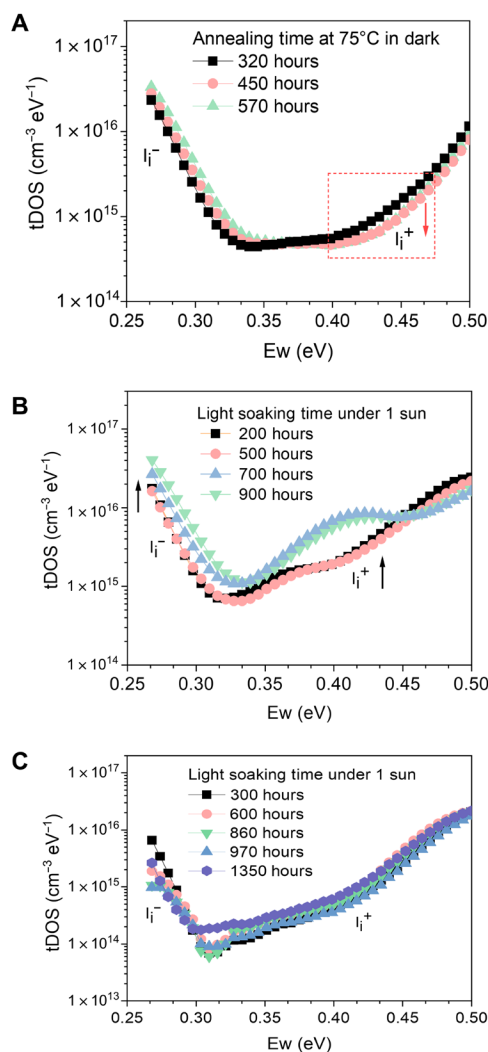


Fig. 5. Evolution of trap density during device operation or annealing. (A) Devices were annealed at 75°C in dark for different hours. (B) The as-made devices were light-soaked under 1 sun for different durations. (C) Devices that were annealed at 75°C in dark for 480 hours to form voids in advance and then light-soaked under 1 sun for different hours. E_w is the demarcation energy determined by $E_w = kT \ln(\omega_0/\omega)$ (where k , T , ω , ω_0 are the Boltzmann's constant, temperature, angular frequency and the attempt-to-escape angular frequency).

illumination, which can also tell whether the formation of voids increases point defect density inside perovskites. To probe the trap energy level and trap density of PSCs, the energetic profile of trap density of states (tDOS) (28, 29) was measured for the devices after light soaking and thermal annealing for different durations. Figure 5A shows tDOS for devices that were annealed at 75°C for different durations in the dark. It was found that the post-annealing of the devices for 350 to 520 hours in the dark caused a reduced trap density with energy depth of ~ 0.4 to 0.48 eV, and there was no obvious change for tDOS for traps at energy depth of ~ 0.25 to 0.35 eV. We have confirmed that these charge traps are caused by positive-charged iodide interstitial (I_i^+) and negative-charged iodide interstitial (I_i^-) (11, 30), respectively, and we speculated the deeper traps are caused by dangling bonds or amorphous regions (22). Therefore, the annealing process that drove the formation of voids did not cause the

increase of iodide interstitials in the perovskite films. The reduction of deep traps can be explained by the reduction of amorphous regions during annealing, which is consistent with morphology study in terms of increased grain size during annealing. Figure 5B shows the tDOS for the as-made devices light-soaked under 1 sun for different durations. It is clearly seen that the trap density in the devices increased after light soaking for 700 hours, especially, the traps with the depth of ~ 0.25 to 0.32 eV and ~ 0.35 to ~ 0.45 eV. The lack of correlation of trap density variation with voids generation again indicates that the formed voids did not cause the device degradation under illumination, and the light-induced device degradation is still dominated by the formation of iodide interstitials in perovskite grains under illumination (11). After post-annealing, the defect generation in the perovskite devices under illumination was markedly slowed down during the followed photostability tests (up to 1000 hours), as shown in Fig. 5C. It is consistent with the improved light stability of devices after post-annealing, pointing out that post-annealing is an effective method to enhance the operational stability of PSCs. It can be explained by the removal of amorphous phase that suppresses ion migration and the ion migration-induced iodide interstitials (3).

DISCUSSION

In conclusion, the generation of voids along the grain boundaries have been observed at the perovskite-substrate interface when the films were exposed to heat and light. The voids were formed by the shrinking of perovskite layers due to the removal of residual DMSO and conversion of amorphous phase to crystalline phase. The post-annealing process, which induced void formation, did not harm the stability of PSCs. It actually decreased the trap density of I_i^+ , which benefited from the removal of amorphous region, leading to enhancement of the light stability of devices. This work points out that the removal of amorphous region from perovskite films is important to enhance the stability of perovskite-based solar cells. Although we have not studied specifically, we speculate that the same conclusion applies to other electronic devices that need stable polycrystalline perovskites.

MATERIALS AND METHODS

Materials

FAI and Methylammonium iodide (MAI) were purchased from GreatCell Solar and used without further purification. PbI_2 (99.99%) was purchased from Tokyo Chemical Industry. 2-Mercaptoethanol (2-ME), DMSO, CsI (99.999%), guanidinium iodide (>99%), formamidinium chloride, phenylethylammonium chloride, zinc chloride, CBH, 2-propanol (IPA; anhydrous, 99.5%), toluene, PTAA (number average molecular weight 7000 to 10,000), and BCP (>96%) were purchased from Sigma-Aldrich. C_{60} (>99.55%) was purchased from Nano-C. Formamidinium hypophosphite was synthesized via a method that was based on, and modified from, a previously published method.

Device fabrication

For the fabrication of $FA_{0.9}Cs_{0.1}PbI_3$ films, ITO glass substrates were first rinsed with soap, deionized water, acetone, and isopropyl alcohol. Then, PTAA solution with the concentration of 2.2 mg/ml in toluene was blade-coated onto the ultraviolet ozone-treated ITO substrates at the speed of 20 mm/s with a gap of 250 μm . $FA_{0.9}Cs_{0.1}PbI_3$ precursor solution was prepared by mixing inks 0.9 M $FAPbI_3$ and 0.1 M $CsPbI_3$ together. The solvent of the perovskite solution is a mixture of 2-ME

and DMSO with a volume ratio of 95:5. The molar percentage of DMSO to Pb is around 70% in precursor solutions. To fabricate high-quality perovskite films, 1.5% formamidinium chloride, 1.0% formamidinium hypophosphite, 0.15% phenylethylammonium chloride, 0.2% zinc chloride, 0.2% CBH, and 0.25% CsI were added to perovskite precursor solution. The $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ films were fabricated via blade coating on PTAA/ITO substrates at room temperature in air at the speed of 20 mm/s with a gap of 300 μm along with N_2 blowing at 20 PSI. After that, the coated $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ films were annealed at 150°C for 3 min to get the perovskite phase.

For the fabrication of $\text{MA}_{0.6}\text{FA}_{0.4}\text{PbI}_3$ films, PTAA solution with the concentration of 3.3 mg/ml in toluene was blade-coated onto the ultraviolet ozone-treated ITO substrates at the speed of 20 mm/s with a gap of 150 μm . $\text{MA}_{0.6}\text{FA}_{0.4}\text{PbI}_3$ precursor solution (1.35 M) was prepared by mixing inks 2.5 M MAPbI_3 and 1.67 M FAPbI_3 together. The solvent of the perovskite solution is a mixture of 2-ME and DMSO. The molar percentage of DMSO to Pb is around 25% in precursor solutions. To fabricate high-quality perovskite films, *n*-dodecylammonium iodide (0.83 mg mL^{-1}), 1- α -phosphatidylcholine (0.27 mg mL^{-1}), 0.14% v/v methylammonium hypophosphite, 4-fluorophenylammonium iodide (1.40 mg mL^{-1}), and 1.5% CBH were added into both precursor solutions as additive. The films were fabricated via blade coating on PTAA/ITO substrates at room temperature in air at the speed of 20 mm/s with a gap of 300 μm along with N_2 blowing at 20 PSI. After that, the coated $\text{MA}_{0.6}\text{FA}_{0.4}\text{PbI}_3$ films were annealed at 120°C for 5 min to get the perovskite phase.

The MAFAPbI_3 films were fabricated via a two-step spin-coating method in the glove box. A PTAA solution with the concentration of 2 mg/ml in toluene was spin-coated onto the ultraviolet ozone-treated ITO substrates at 5000 rpm for 30 s, followed by an annealing process at 100°C for 10 min. To fabricate films with or without DMSO, 400 mg of PbI_2 was dissolved in 1 ml of DMF, or 400 mg of PbI_2 was dissolved in DMF:DMSO mixing solvent (v:v 4:1). Then, a PbI_2 solution was first spin-coated onto PTAA at 1500 rpm for 30 s and then annealed at 70°C for 5 min. After cooling down to room temperature, a solution of FAI:MAI:MACl (44 mg:4 mg:4.4 mg in 1 ml of IPA) was spin-coated onto the PbI_2 at 1500 rpm for 30 s. The perovskite films were annealed at 150°C for 15 min.

The complete device was assembled by sequentially thermal evaporating C_{60} (30 nm), dynamically spin coating BCP (0.5 mg/ml in IPA at 5000 rpm), and thermal evaporating Cu electrodes (150 nm). The active area of all the solar cells reported in this work was 8 mm^2 , defined by the overlap of ITO and Cu electrode. The complete devices were encapsulated by epoxy in air.

Characterizations

Electronic characterizations

The *J*-*V* characteristics of solar cells were measured by a Xenon lamp-based solar simulator (Oriel Sol3A, Class AAA Solar Simulator) with power of 100 mW cm^{-2} under simulated illumination of AM 1.5 G by a silicon reference cell (Newport 91150 V-KG5). The scanning scan rate is 0.1 V s^{-1} , and the scanning direction was from positive to negative voltages. The measurement was conducted at room temperature, while the device might have temperature above room temperature due to the light heating effect. There was no preconditioning before measurement. The active area of all the solar cells reported in this work was 8 mm^2 , which is defined as the areas have both cathode and anode. The tDOS of solar cells was derived from the frequency-dependent capacitance and voltage-dependent

capacitance by the thermal admittance spectroscopy measurement performed with an LCR meter (Agilent E4980A).

Structural and composition characterizations

SEM images were taken on Hitachi 4700. The accelerating voltage of the electron beam is 2 kV, and the current is 10 μA . XRD measurements were carried out with a Rigaku SmartLab diffractometer using $\text{Cu K}\alpha$ radiation (a wavelength of 1.5418 Å). ^1H NMR spectra were acquired with a Bruker AVANCE III 600-MHz NMR spectrometer. The NMR liquid samples were prepared as follows: First, the perovskite films were blade-coated onto glass substrates and then annealed on a hot plate for different durations at 150°C. After that, the perovskite films were carefully scraped off glass substrates with a stainless-steel blade, and the perovskite powers were quickly transferred to NMR tubes filled with deuterium oxide, which were then shaken thoroughly before the characterization.

SUPPLEMENTARY MATERIALS

Supplementary material for this article is available at <https://science.org/doi/10.1126/sciadv.abo5977>

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